

Lab 1: Python Basics, File Handling, NumPy, and Pandas

Duration: 1 Hour

Objecti

Part 1: Python Basics - Variables, Functions, and Loops

1. Define a Function to Calculate Factorial

Fill in the

```
def factorial(n):
```

```
    """
```

```
# [OK] Test Case
```

```
def t
```

```
test_factorial()
```

Output:

[OK] All test cases passed!

2. Reverse a String

Complete

```
def reverse_string(s):
```

```
    """
```

```
# [OK] Test Case
```

```
def t
```

```
test_reverse_string()
```

Output:

[OK] All test cases passed!

3. Find Even Numbers in a List

Complete

```
def find_even_numbers(lst):
```

```
    """
```

```
# [OK] Test Case
```

```
def t
```

```
test_find_even_numbers()
```

Output:

[OK] All test cases passed!

Part 2: File Handling

4. Writing and Reading a File

Fill in the

```
# Writing to a file
```

```
def w
```

```
# Reading from a file
```

```
def r
```

```
# [OK] Test Case
```

```
def t
```

```
write_to_file(filename, names)
```

```
co
```

```
assert content == ["Alice\n", "Bob\n", "Charlie\n"], "Test case failed!"
```

```
pr
```

```
test_file_handling()
```

Output:

[OK] All test cases passed!

5. Writing and Reading a JSON File

Fill in the

```
import json
```

```
# Writing to a JSON file
```

```
def w
```

```
# Reading from a JSON file
```

```
def r
```

```
# [OK] Test Case
```

```
def test
```

```
    write_json(filename, data)
```

```
    l
```

```
    assert loaded_data == data, "Test case failed!"
```

```
    p
```

```
test_json_handling()
```

Output:

[OK] All test cases passed!